8 - lesson report

Mavzu: For loop + encoding, scaling

Bugungi darsda maʼlumotlarni qayta ishlashda muhim boʼlgan ikki tushuncha oʼrganildi. Encoding va scaling. Bu jarayonlar koʼpincha mashinaviy oʼrganish modellarini tayyorlashdan oldin maʼlumotlarni mos formatga keltirish uchun ishlatiladi. Dars davomida ushbu amallarni for loop yordamida qanday bajarish mumkinligini koʼrib chiqildi.

Encoding - bu katekorig (matnli va toifaviy) maʼlumotlarni sonli koʼrinishga oʼtkazish jarayonidir. Chunki koʼpgina AI va ML algoritmlari faqat sonli maʼlumotlar bilan ishlaydi.

For loop yordamida encoding qilishda: Har bir element ketma-ket koʼrib chiqiladi, har bir kategoriyaga mos son biriktiriladi, natijada yangi sonli roʼyxat hosil boʼladi.

Misol: Agar ranglar roʼyxati boʼlsa: ["red", "blue", "green", "blue"] ularni quyidagicha kodlash mumkin : red => 0, blue +> 1, green => 2, Natija: [0,1,2,1]

```python
from sklearn.preprocessing import LabelEncoder
```

[13] ✓ 0.0s Python

```python
encoder = LabelEncoder()
```

[14] ✓ 0.0s Python

```python
for col in df.columns:
    if df[col].dtype=='object':
        if df[col].nunique()<=5:
            dummies=pd.get_dummies(df[col], prefix=col, dtype=int)
            df=pd.concat([df.drop(columns=col), dummies], axis=1)

        else:
            df[col]=encoder.fit_transform(df[col])
```

[15] ✓ 0.0s Python

```python
df.head()
```

[16] ✓ 0.0s Python

	Make	Model	Body	Mileage_km	Price	Year	Country	Fuel_Type	Fuel_Consumption_l	Gears	...
0	0	124	0	98000	16900	2020.0	5	6	6.700000	5.899024	...
1	0	124	5	91500	12500	2017.0	5	6	6.000000	5.000000	...
2	0	124	0	40000	17990	2015.0	5	6	6.500000	5.000000	...
3	0	100	0	133000	9300	2008.0	5	6	6.500000	5.000000	...
4	0	124	0	61019	14990	2021.0	1	6	6.727151	6.000000	...

Bu jarayon for loop orqali har bir elementni tekshirib, mos qiymatni qo'shish orqali bajariladi.

For loop + Scaling

Scaling - bu sonli ma'lumotlarni bir xil diapazonga keltirish jarayonidir. bu modelning samarali ishlashi uchun myhim, chunki turli kattalikdagi sonlar modelga salbiy ta'sir qilishi mumkin.

ENg ko’p ishlatiladigan scaling usullari: MinMax scaling (0dan 1gacha), Standard Scaling (o’rtacha = 0, dispersiya = 1)

For loop yordamida scaling qilishda: Har bir son ustida matematik formula qo’llaniladi, natijalar yangi ro’yxatga yoziladi.

Misol: (MinMax) Agar ma’lumotlar [10,20,30] Min = 10, Max = 30 Formula (x - min) / (max - min) Natija: [0, 0.5, 1]

Robust Scaler ma’lumotlarni median va interkvartil oraliq (IQR) asosida masshtablash usulidir. Oddiy scaling usullari (masalan, Min-Max yoki Standard Scaler) o’rtacha qiymat va minimum-maksimumlarga tayanadi, shu sababli outlierlar ularni buzib yuborishi mumkin. Robust Scaler esa median va kvartillarni ishlatgani uchun ekstremal qiymatlarga kamroq sezgir.

Formula:(x − median) / IQR

Median - ma’lumotlarni o’rta qiymati, IQR (Interquartile Range) = Q3 -Q1, Q1 - 25 foizli kvartil, Q3 - 75 foizli kvartil,

Robust scaler quyidagi holatlarda ayniqsa foydali:

Datasetda outlierlar ko'p bo'lsa, ma'lumotlar notekis taqsimlangan bo'lsa, narx, daromad, masofa kabi keskin farq qiluvchi qiymatlar mavjud bo'lsa.

```
[17]  from sklearn.preprocessing import StandardScaler, MinMaxScaler, RobustScaler
      ✓ 0.0s

[18]  num_col = df.select_dtypes(include=['int64', 'float64']).columns.drop('Year')
      ✓ 0.0s

[19]  num_col
      ✓ 0.0s

...   Index(['Mileage_km', 'Price', 'Fuel_Consumption_l', 'Gears', 'Power_hp',
            'Engine_Size_cc', 'Cylinders', 'Seats', 'Doors', 'Previous_Owners'],
           dtype='object')

[20]  scaler=StandardScaler()
      ✓ 0.0s

[21]  df['Mileage_km']=scaler.fit_transform(df[['Mileage_km']])
      ✓ 0.0s

[22]  df.head()
      ✓ 0.0s
```

	Make	Model	Body	Mileage_km	Price	Year	Country	Fuel_Type	Fuel_Consumption_l
0	0	124	0	0.064576	16900	2020.0	5	6	6.700000
1	0	124	5	0.044366	12500	2017.0	5	6	6.000000
2	0	124	0	-0.115757	17990	2015.0	5	6	6.500000
3	0	100	0	0.173398	9300	2008.0	5	6	6.500000
4	0	124	0	-0.050405	14990	2021.0	1	6	6.727151

```
scaler=MinMaxScaler()
```
[23] ✓ 0.0s

```
df['Price']=scaler.fit_transform(df[['Price']])
```
[24] ✓ 0.0s

```
df.head()
```
[25] ✓ 0.0s

	Make	Model	Body	Mileage_km	Price	Year	Country	Fuel_Type	Fuel_Consumptior
0	0	124	0	0.064576	0.001690	2020.0	5	6	6.7000
1	0	124	5	0.044366	0.001250	2017.0	5	6	6.0000
2	0	124	0	-0.115757	0.001799	2015.0	5	6	6.5000
3	0	100	0	0.173398	0.000930	2008.0	5	6	6.5000
4	0	124	0	-0.050405	0.001499	2021.0	1	6	6.7271

5 rows × 31 columns

```
scaler=RobustScaler()
```
[26] ✓ 0.0s

```
df['Previous_Owners']=scaler.fit_transform(df[['Previous_Owners']])
```
[27] ✓ 0.0s

```
df.head()
```
[28] ✓ 0.0s

	Make	Model	Body	Mileage_km	Price	Year	Country	Fuel_Type	Fuel_Consumptior
0	0	124	0	0.064576	0.001690	2020.0	5	6	6.7000
1	0	124	5	0.044366	0.001250	2017.0	5	6	6.0000

Bugungi darsda for loop yordamida ma'lumotlarni encoding va scaling qilish usullari o'rganildi. bu bilimlar mashinaviy o'rganish modellarini yaratishda ma'lumotlarni tayyorlash bosqichida juda muhim hisoblanadi. For loop

orqali bu jarayonlarni tushunish dasturlash asoslarini mustahkamlashga ham yordam beradi.